

# NIPUN GAHANE

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## EDUCATION

**B.TECH – Computer Science Engineering**, SRM Institute of Science and Technology Apr 2025  
Specialization in **Software Engineering**  
Relevant Coursework: **Data Structures & Algorithms, Database Management Systems, Machine Learning**  
GPA: 8.2 / 10.0

## EXPERIENCE

**Standard Chartered GBS - Bangalore** Jul 2025 – Present  
Technology - *Apprentice (Machine Learning & AI Focus)*

- Contributing to the High-Level Design (**HLD**) and architectural planning of an enterprise Retrieval-Augmented Generation (**RAG**) chatbot intended to automate internal banking queries.
- Collaborating with cross-functional teams to evaluate **Large Language Model (LLM)** capabilities, vector database solutions, and integration strategies for future production deployment.
- Built and deployed a scalable **Spring Boot** API to facilitate high-throughput data remediation for downstream ML processes, integrating **JPA** and automated **CI/CD** workflows via **Azure Pipelines**.

## PROJECTS

**Finance RAG App (SEC 10-K Q&A)** Mar 2026  
Built an end-to-end financial **RAG** platform to ingest SEC filings, generating grounded answers with source citations via a **FastAPI** interface. Implemented hybrid retrieval using **ChromaDB** vector search and BM25 sparse search with RRF fusion, improving Recall@5 by ~52% versus a vector-only baseline. **Tech:** Python, **LangChain**, **Ollama**, **ChromaDB**, **PyTorch**.

**Efficient LLM Finetuning (Google Gemma 2B)** Oct 2025  
Executed parameter-efficient finetuning (**PEFT**) on the Google Gemma 2B model using **LoRA** and 4-bit quantization. Trained the **LLM** on specialized datasets to adapt generative capabilities. **Tech:** **PyTorch**, **Hugging Face** (Transformers, TRL), **LoRA**.

**End-to-End MLOps Pipeline** Feb 2024  
Designed and automated an **MLOps** pipeline to manage the complete model lifecycle for predictive analytics, achieving an R-squared of 88.28% while ensuring continuous integration of data processing steps. **Tech:** **TensorFlow**, **scikit-learn**, **Flask**.

**Real-Time NLP Spam Detection API** Jan 2024  
Deployed a low-latency **Natural Language Processing (NLP)** detection API utilizing a **PyTorch** neural network, effectively processing streaming text data. **Tech:** **PyTorch**, **NLTK**, **REST APIs**.

**Cloud-Native Scalable Backend System** Mar 2026  
Engineered a cloud-native backend using **Spring Boot** and **MongoDB Atlas**, deploying microservices via **Docker** and **AWS ECS (Fargate)**. Integrated **GitHub Actions CI/CD** to ensure robust delivery for scalable data applications. **Tech:** **Java**, **Spring Boot**, **AWS**, **Docker**.

**Deepfake Video Detection (DART)** Dec 2023  
Constructed a computer vision model achieving 92% accuracy using a **TensorFlow CNN-RNN** architecture with a real-time interface. **Tech:** **TensorFlow**, **OpenCV**, **CNNs**.

## TECHNICAL SKILLS

|                      |                                                                    |
|----------------------|--------------------------------------------------------------------|
| Languages            | Python (Proficient), Java, SQL                                     |
| AI / ML Technologies | PyTorch, TensorFlow, scikit-learn, Hugging Face, LangChain, OpenCV |
| Developer Tools      | Git, Docker, AWS, Azure, MLflow, Jupyter, CI/CD                    |
| Core Competencies    | Machine Learning, Generative AI (LLMs/RAG), NLP, MLOps, API Design |

## ACHIEVEMENTS & ACTIVITIES

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- **Competitive Programming:** Solved 300+ problems on [LeetCode](#), demonstrating strong proficiency in data structures, algorithms, and complex problem-solving.
- **Machine Learning Specialization:** Completed the [DeepLearning.AI certification](#) focusing on advanced neural networks and ML architectures.